The electromagnetic spectrum

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electromagnetic wavelength, frequency and propagation.

SUCCESS CRITERIA - I can...

1. Use c=fλ in vacuum.
2. Order spectral regions by wavelength.
3. Distinguish electromagnetic from mechanical waves.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “electromagnetic” and “spectrum” to explain your answer. Sentence starters are optional.

electromagnetic: involving coupled electric and magnetic fields

spectrum: a range arranged by frequency or wavelength

vacuum: space with negligible matter

01 Warm-up: bring back an earlier idea

An echo travels **100 m** on its full out-and-back path. **Find the wall’s distance**.

Word help: echo = a sound received after reflection

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: c = f λ; c = **3.0×10^{8} m/s**.
Electromagnetic waves in vacuum. Frequency identifies spectral region; mechanical sound needs a medium.

Worked reasoning: In vacuum, all electromagnetic waves travel at c=**3×10^{8} m/s**. A frequency of **10^{8} Hz** gives wavelength **3 m**. Higher frequency means shorter wavelength at the same speed.

Your turn: Draw two waves with a common speed but different wavelengths; **label which has higher frequency**.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Order radio, visible and X-rays from longest to shortest wavelength **0**. Which has greatest frequency in vacuum?

Sentence starter: I notice ... . This supports ... because ... .

Radio	IR	Visible	UV	X-ray

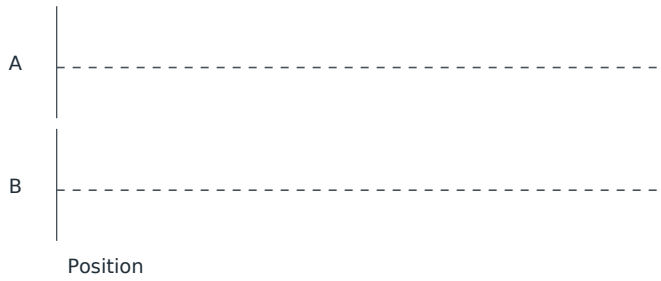
Label increasing frequency and wavelength.

Marzano check - circle: ① ② ③

04 Apply the physics

Use a slow-motion wave simulation with speed **12 m/s** and frequency **3 Hz**. **Calculate wavelength**. Then double frequency while holding speed fixed. **Sketch both patterns over the** same distance and **label wavelengths**. The simulated speed is not the speed of light.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

Why can sunlight cross space but ordinary sound from a speaker cannot cross a vacuum?

Sentence starter: I think … . I can point to …, which shows …, so … .

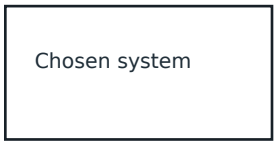
Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A system absorbs **80 J** heat and does **40 J** work. **Find its internal-energy change** using $\Delta U = Q - W_{by}$.
Label energy entering, leaving and stored.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.



Stored energy change: _____ J

Marzano check - circle: ① ② ③

07 Exit check

For EM waves in vacuum, does changing color change c? What does change?

Sentence starter: I notice … . This supports … because … .

Radio	IR	Visible	UV	X-ray

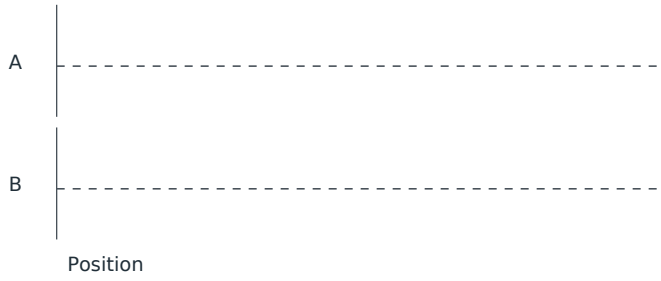
Label increasing frequency and wavelength.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Design two frequency-wavelength pairs with product **3×10^{8} m/s**, using powers of **10**.

Sentence starter: My design includes … . It does what is needed by … .



Marzano check - circle: ① ② ③